

RIT Client REST API ^{1.0.3}

[Base URL: localhost:9999/v1]

<https://rit.306w.ca/RIT-REST-API/1.0.3/swagger.yaml>

Overview

The Rotman Interactive Trader (RIT) allows users to query market data and submit trading instructions through a REST API, as well as through a Microsoft Excel VBA API. The REST API is further categorized into two approaches: the Client REST API and the DMA (Direct Market Access) REST API. While these two approaches use the same REST API type, they differ in the connection endpoints. The Client REST API connects to the RIT Client, whereas the DMA REST API connects directly to the RIT Instructor App (Server). This webpage aims to provide detailed information on the **Client REST API**. If you are looking for detailed information about Excel VBA API, please click [here](#). For the DMA REST API, please click [here](#).

The Client REST API is a simple interface for interacting with the Rotman Interactive Trader (RIT) Client. RIT Client must be running to access this API. Responses are always [JSON objects](#). Successful requests always return with HTTP code **200**. Unsuccessful responses have HTTP error codes between **400** and **500**. Most programming languages have ways to interact with this kind of API. Some resources:

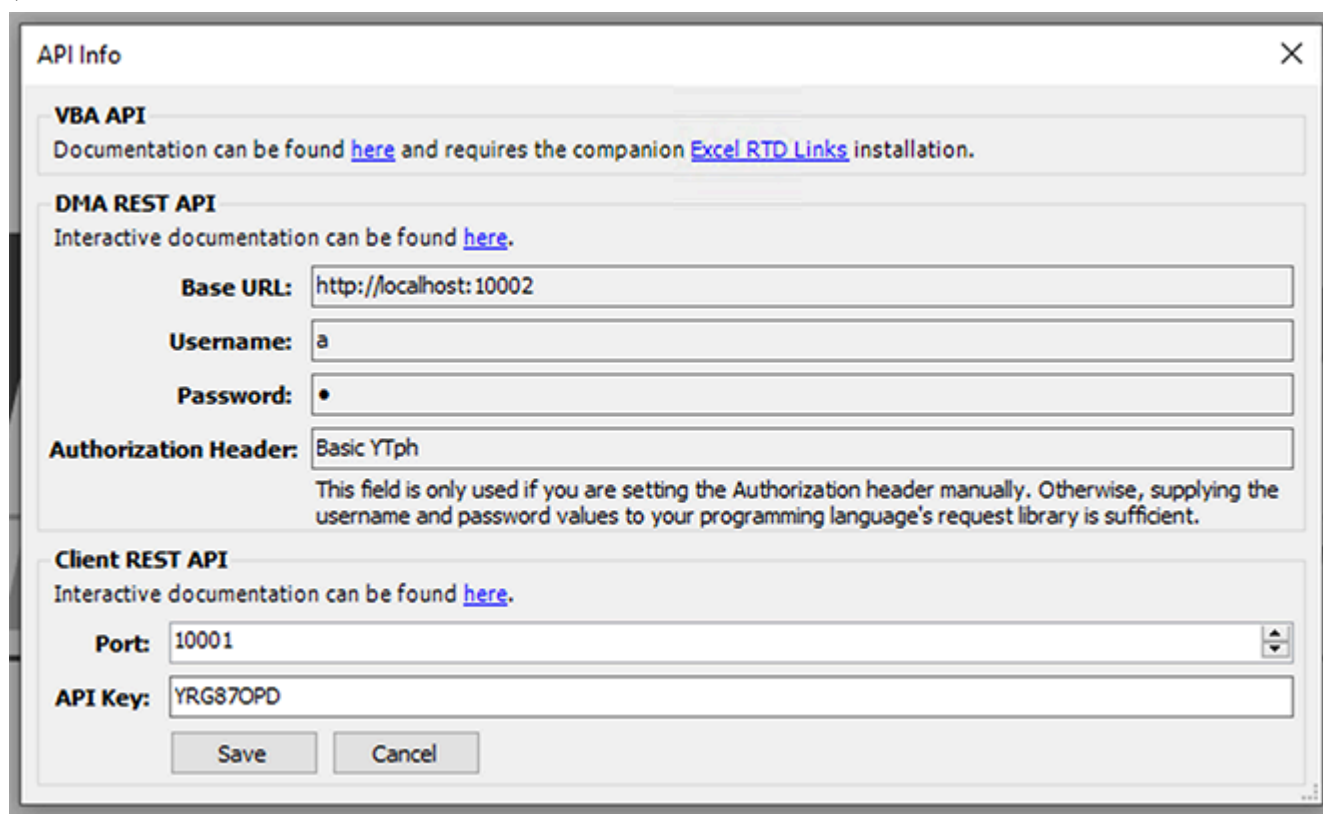
- MATLAB <https://www.mathworks.com/help/mps/restful-api-and-json.html?requestedDomain=www.mathworks.com>
- C# <https://docs.microsoft.com/en-us/aspnet/web-api/overview/advanced/calling-a-web-api-from-a-net-client>
- Python <https://realpython.com/blog/python/api-integration-in-python/>
- Node <https://www.npmjs.com/package/request>
- VBA <https://stackoverflow.com/questions/19553476/how-to-make-rest-call-with-vba-in-excel> along with <https://stackoverflow.com/questions/19360440/how-to-parse-json-with-vba-without-external-libraries>

Configuration



The **API** icon indicates whether an API access is enabled. If there are any issues with starting the API, the icon will be **red** and clicking on it will reveal the associated error message. If the API is turned off, the icon will be greyed out. This icon applies to both Excel VBA API and REST API. Similarly, the **API Orders** icon indicates whether orders can be submitted via API.

Users can click on this **API** icon in order to access more detailed information on interacting with the API.



The screenshot shows a dialog box titled "API Info" with a close button (X) in the top right corner. It contains three sections:

- VBA API**: Documentation can be found [here](#) and requires the companion [Excel RTD Links](#) installation.
- DMA REST API**: Interactive documentation can be found [here](#).
 - Base URL:**
 - Username:**
 - Password:**
 - Authorization Header:**
This field is only used if you are setting the Authorization header manually. Otherwise, supplying the username and password values to your programming language's request library is sufficient.
- Client REST API**: Interactive documentation can be found [here](#).
 - Port:**
 - API Key:**

At the bottom are "Save" and "Cancel" buttons.

The REST API by default runs on port **9999** but that can be modified here. The API key (initially autogenerated) can also be changed here. The API will not be accessible without an API key configured. Clicking through the Authorize button below will initialize this page with the proper port and API key settings for a particular client instance and is the recommended way to start exploring this API.

Authentication

The API key can be found by clicking on the **API** icon in the status bar once logged in. This key must be included in every request to the RIT API, either through the **X-API-Key** header or through the **key** query parameter. Requests without a matching key will be rejected with HTTP code **401**.

Rate Limiting

Certain endpoints are rate-limited (indicated in the endpoint description). When a request is rejected due to rate-limiting, the response will have an HTTP error code of **429**. In addition, the response will have a **Retry-After** header and **wait** body parameter which indicates the number of seconds to wait before attempting the request again.

Note that in addition to the global rate limit on requests, securities may also have additional, more restrictive rate limits which apply to requests for those securities. The **Retry-After** / **wait** times in those cases correspond to the more restrictive rate limits. Requests for other securities with less restrictive rate limits may still be successful.

Generally, most of the default RIT cases will have a very 'loose' rate-limiting otherwise changed by instructors. Therefore, it is not expected that users will often experience this response from rate-limiting.

The Excel VBA API treats rate-limiting in a different way and instead queues any requests which exceed the rate limit (and executes them over time at the rate limit). If the Excel VBA API is used concurrently

with this Client REST API, endpoints which are rate-limited will reject requests if there are any queued requests. It is NOT advised to run both APIs simultaneously.

Additional Notes

- Tickers are case-sensitive and should match what is displayed in the RIT Client.

Schemes

HTTP

Authorize



default

GET /case Gets information about the current case.



GET /trader Gets information about the currently signed in trader.



GET /limits Trading limits for the current case.



GET /news Gets the most recent news.



GET /assets Gets a list of available assets.



GET /assets/history Gets the activity log for assets.



GET /securities Gets a list of available securities and associated positions.



GET /securities/book Gets the order book of a security.



GET /securities/history Gets the OHLC history for a security.



GET /securities/tas Gets time & sales history for a security.



GET /orders Gets a list of all orders.



POST `/orders` Insert a new order.



GET `/orders/{id}` Gets the details of a specific order.



DELETE `/orders/{id}` Cancel an open order.



GET `/tenders` Gets a list of all active tenders.



POST `/tenders/{id}` Accept the tender.



DELETE `/tenders/{id}` Decline the tender.



GET `/leases` List of all assets currently being leased or being used.



POST `/leases` Lease or use an asset.



GET `/leases/{id}` Gets the details of a specific lease.



POST `/leases/{id}` Use a leased asset.



DELETE `/leases/{id}` Unlease an asset.



POST `/commands/cancel` Bulk cancel open orders.



Models



```
Error {
  code*      string
  message    string
}
```

```
CaseStatus string
Enum:
  Array [ 3 ]
```

OrderType string

Enum:

Array [2]

SecurityType string

Enum:

Array [12]

AssetType string

Enum:

Array [6]

TickerQuantity {

ticker	string <i>example: CL</i>
quantity	number <i>example: 10</i>

}

TickerPrice {

ticker	string <i>example: CL</i>
price	number <i>example: 9.99</i>

}

OrderAction string

Enum:

Array [2]

OrderStatus string

Enum:

Array [3]

Order {

order_id	integer <i>example: 1221</i>
period	integer <i>example: 1</i>
tick	integer <i>example: 10</i>
trader_id	string <i>example: trader49</i>
ticker	string <i>example: CRZY</i> Case sensitive.
type	string <i>example: LIMIT</i>

```

    Enum:
        Array [ 2 ]
    quantity      number
                  example: 100
    action         string
    Enum:
        Array [ 2 ]
    price          number
                  example: 14.21
                  Will be null if type is LIMIT.
    quantity_filled number
                  example: 10
                  Always less than or equal to quantity.
    vwap           number
                  example: 14.21
                  Volume-weighted average price for transacted quantity. null
                  if quantity_filled is 0.
    status         string
    Enum:
        Array [ 3 ]
}

```

```

AssetLease {
    id      integer
           Lease id.
    ticker  string
    type    string
           example: CONTAINER
    Enum:
        Array [ 6 ]
    start_lease_period integer
    start_lease_tick   integer
    next_lease_period  integer
    next_lease_tick    integer
    convert_from       [...]
    convert_to         [...]
    containment_usage  integer
                     Currently utilized capacity for CONTAINER type assets.
}

```

```

SuccessResult {
    description: Placeholder result to indicate an operation was successful.
    success     boolean
               example: true
}

```

VALID

{...}